

**SecureSmart Intern: AI-Powered Internship  
Recommendation & Scam Detection Platform**

**A PROJECT REPORT**

*Submitted by*

**ZALA PRAGANABA [23BECE30402]**

**TRIVEDI DIYA ASHISH [23BECE30374]**

**VARMA RIDDHI HARESH [23BECE30388]**

**VISHVA PATEL [224 SBECE30053]**

*In fulfillment for the award of the degree*

*of*

**BACHELOR OF ENGINEERING**

*in*

**COMPUTER ENGINEERING**



**LDRP Institute of Technology and Research, Gandhinagar**

**Kadi Sarva Vishwavidyalaya**

**February, 2026**

# ***LDRP INSTITUTE OF TECHNOLOGY AND RESEARCH***

## ***GANDHINAGAR***

CE-IT Department



## **CERTIFICATE**

This is to certify that the Project Work entitled **“SecureSmart Intern: AI-Powered Internship Recommendation & Scam Detection Platform”** has been carried out by **Zala Pragnaba (23BECE30402)** under my guidance in fulfilment of the degree of Bachelor of Engineering in Computer Engineering department Semester-6 of Kadi Sarva Vishwavidyalaya during the academic year 2025-2026.

**Prof. Manisha Chaudhary**

Internal Guide

LDRP ITR

**Dr. Ashish Patel**

Head of the Department

LDRP ITR

# ***LDRP INSTITUTE OF TECHNOLOGY AND RESEARCH***

## ***GANDHINAGAR***

CE-IT Department



## **CERTIFICATE**

This is to certify that the Project Work entitled **“SecureSmart Intern: AI-Powered Internship Recommendation & Scam Detection Platform”** has been carried out by **Trivedi Diva Ashish (23BECE30374)** under my guidance in fulfilment of the degree of Bachelor of Engineering in Computer Engineering department Semester-6 of Kadi Sarva Vishwavidyalaya during the academic year 2025-2026.

**Prof. Manisha Chaudhary**

Internal Guide

LDRP ITR

**Dr. Ashish Patel**

Head of the Department

LDRP ITR

**LDRP INSTITUTE OF TECHNOLOGY AND RESEARCH**  
**GANDHINAGAR**

CE-IT Department



**CERTIFICATE**

This is to certify that the Project Work entitled **“SecureSmart Intern: AI-Powered Internship Recommendation & Scam Detection Platform”** has been carried out by **Varma Riddhi Haresh (23BECE30388)** under my guidance in fulfilment of the degree of Bachelor of Engineering in Computer Engineering department Semester-6 of Kadi Sarva Vishwavidyalaya during the academic year 2025-2026.

**Prof. Manisha Chaudhary**

Internal Guide

LDRP ITR

**Dr. Ashish Patel**

Head of the Department

LDRP ITR

**LDRP INSTITUTE OF TECHNOLOGY AND RESEARCH  
GANDHINAGAR**

**CE-IT Department**



**CERTIFICATE**

This is to certify that the Project Work entitled **“SecureSmart Intern: AI-Powered Internship Recommendation & Scam Detection Platform”** has been carried out by **Vishva Patel** (**224 SBECE30053**) under my guidance in fulfilment of the degree of Bachelor of Engineering in Computer Engineering department Semester-6 of Kadi Sarva Vishwavidyalaya during the academic year 2025-2026.

**Prof. Manisha Chaudhary**

Internal Guide

LDRP ITR

**Dr. Ashish Patel,**

Head of the Department

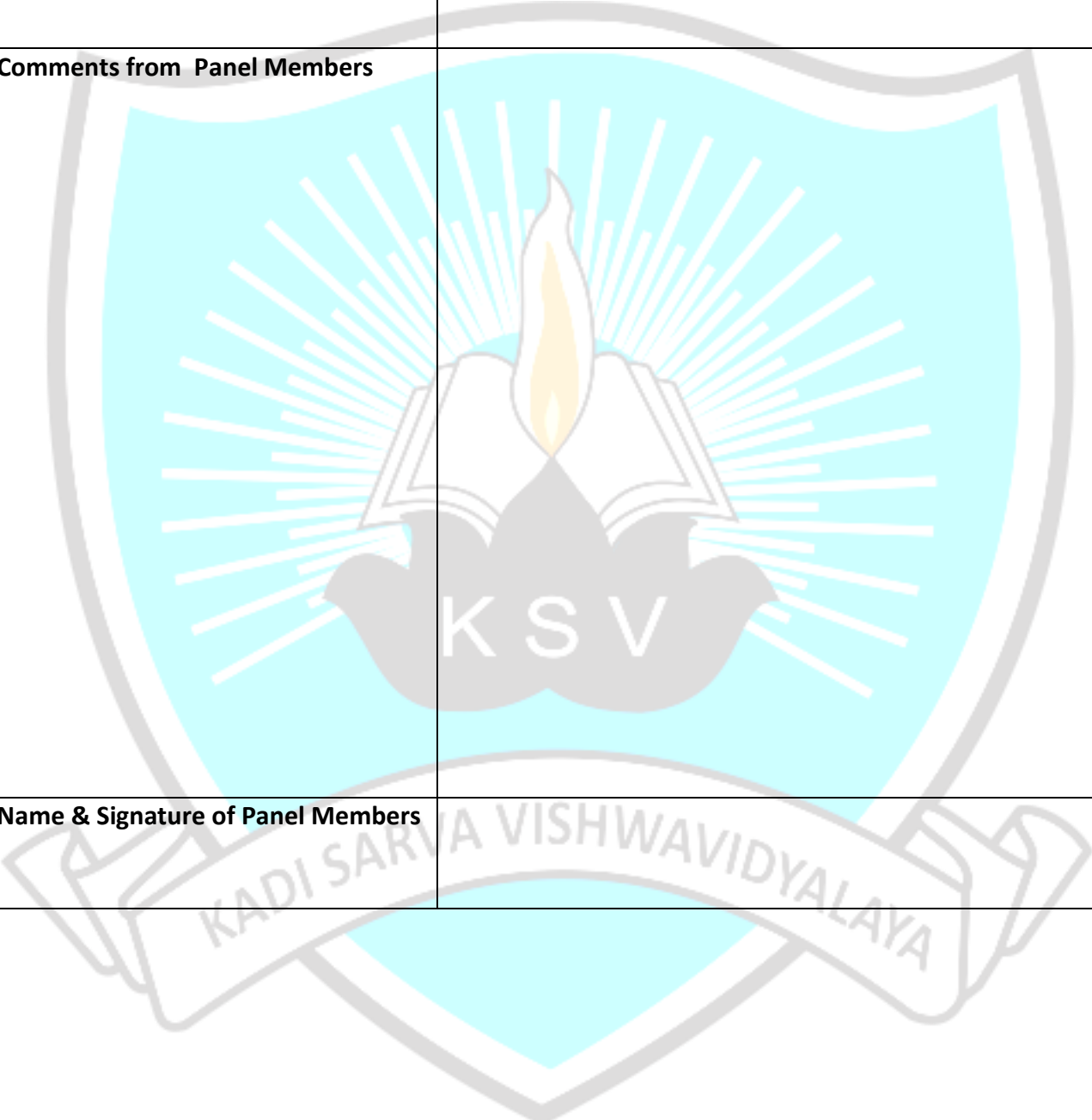
LDRP ITR

### Presentation-I for Project-I

|                                                  |  |
|--------------------------------------------------|--|
| <b>1. Name &amp; Signature of Internal Guide</b> |  |
| <b>2. Comments from Panel Members</b>            |  |
| <b>3. Name &amp; Signature of Panel Members</b>  |  |

## Presentation-II for Project-I

|                                       |  |
|---------------------------------------|--|
| 1. Name & Signature of Internal Guide |  |
| 2. Comments from Panel Members        |  |
| 3. Name & Signature of Panel Members  |  |





## **ACKNOWLEDGEMENT**

We would like to express my profound gratitude to Prof. Manisha Chaudhary Ma'am who gave us the golden opportunity and the guidance to do this wonderful project, which also helped me in doing a lot of research and we came to know about so many new things we are really thankful to them.

**ZALA PRAGNABA [23BECE30402]**

**TRIVEDI DIYA ASHISH [23BECE30374]**

**VARMA RIDDHI HARESH [23BECE30388]**

**VISHVA PATEL [224 SBECE30053]**



## ABSTRACT

*In today's competitive academic and professional environment, students often face difficulty in discovering genuine internship opportunities while avoiding fraudulent postings. SecureSmart Intern is an AI-powered web platform designed to address these challenges by providing personalized internship recommendations and detecting potentially fake internships. The system allows students to create detailed skill-based profiles, explore suitable internships, track applications, and receive guidance on missing skills to enhance their employability. The AI-assisted recommendation engine calculates skill match percentages, ensuring that students find internships aligned with their expertise.*

*The platform also incorporates a scam detection module that evaluates internship listings for suspicious characteristics such as unrealistic stipends, vague company details, or upfront payment requests. Companies can post internships, manage applications, and view applicants through a dedicated dashboard. By combining personalized internship recommendations with proactive fraud detection, SecureSmart Intern provides a safer and more efficient way for students to access genuine opportunities while supporting companies in finding skilled candidates.*

## TABLE OF CONTENTS

| NO       | CHAPTER NAME                            | PAGE NO. |
|----------|-----------------------------------------|----------|
|          | <b>Acknowledgement</b>                  | viii     |
|          | <b>Abstract</b>                         | ix       |
|          | <b>List Of Figures</b>                  | xiii     |
|          | <b>List Of Tables</b>                   | xiv      |
|          | <b>Table Of Contents</b>                | x        |
| <b>1</b> | <b>Introduction</b>                     | 2        |
|          | 1.1 Introduction                        | 2        |
|          | 1.2 Aims and Objective of the Work      | 2        |
|          | 1.3 Brief Literature Review             | 2        |
|          | 1.4 Problem Definition                  | 3        |
|          | 1.5 Plan Of the Work                    | 3        |
| <b>2</b> | <b>Technology and Literature Review</b> | 5        |
|          | 2.1 Technology Overview                 | 6        |
|          | 2.2 Literature Review                   | 6        |
|          | 2.3 Research Gap and Motivation         | 7        |
| <b>3</b> | <b>System Requirement Study</b>         | 8        |
|          | 3.1 User Characteristics                | 9        |
|          | 3.1.1 Student Users                     | 9        |
|          | 3.1.2 Company Users                     | 9        |
|          | 3.1.3 Administrator                     | 10       |
|          | 3.2 Software and Hardware Requirements  | 10       |
|          | 3.2.1 Software Requirements             | 10       |

|          |                                  |    |
|----------|----------------------------------|----|
|          | 3.2.2 Hardware Requirements      | 10 |
|          | 3.3 Assumptions and Dependencies | 11 |
|          | 3.3.1 Assumptions                | 11 |
|          | 3.3.2 Dependencies               | 11 |
| <b>4</b> | <b>System Diagrams</b>           | 12 |
|          | 4.1 Class Diagram                | 13 |
|          | 4.2 Use Case Diagram             | 15 |
|          | 4.3 Entity Relationship Diagram  | 17 |
|          | 4.4 Sequence Diagram             | 18 |
|          | 4.5 Activity Diagram             | 20 |
|          | 4.6 Data Flow Diagram            | 22 |
|          | 4.6.1 Context-Level DFD          | 22 |
|          | 4.6.2 First-Level DFD            | 23 |
|          | 4.6.3 Second-Level DFD           | 24 |
| <b>5</b> | <b>Data Dictionary</b>           | 26 |
|          | 5.1 Student Entity               | 28 |
|          | 5.2 Company Entity               | 29 |
|          | 5.3 Application and Report       | 29 |
|          | 5.4 Internship Posting           | 30 |
| <b>6</b> | <b>Snapshots</b>                 | 32 |
|          | 6.1 Website Homepage             | 32 |
|          | 6.2 Student                      | 32 |
|          | 6.2.1 Student Registration       | 32 |
|          | 6.2.2 Student Login              | 33 |

|          |                                          |           |
|----------|------------------------------------------|-----------|
|          | 6.2.3 Student Homepage                   | 33        |
|          | 6.2.4 Student Profile                    | 34        |
|          | 6.2.5 Recommendations                    | 34        |
|          | 6.2.6 Report                             | 35        |
|          | 6.3 Company                              | 36        |
|          | 6.3.1 Company Registration               | 36        |
|          | 6.3.2 Company Login                      | 36        |
|          | 6.3.3 Company Homepage                   | 37        |
|          | 6.3.4 Company Post Internships           | 37        |
| <b>7</b> | <b>Result, Discussion and Conclusion</b> | <b>38</b> |
|          | 7.1 Result                               | 39        |
|          | 7.2 Discussion                           | 39        |
|          | 7.3 Conclusion                           | 39        |
| <b>8</b> | <b>Bibliographies</b>                    | <b>40</b> |
|          | 8.1 References                           | 41        |

**LIST OF FIGURES**

| <b>NO.</b> | <b>NAME</b>                                        | <b>PAGE NO.</b> |
|------------|----------------------------------------------------|-----------------|
| 1          | Flowchart of Project Development plan              | 4               |
| 2          | Class Diagram                                      | 14              |
| 3          | Use Case Diagram                                   | 16              |
| 4          | Entity Relationship Diagram                        | 17              |
| 5          | Sequence Diagram                                   | 19              |
| 6          | Activity Diagram                                   | 21              |
| 7          | Context Level DFD (Level 0)                        | 22              |
| 8          | First Level DFD (Level 1)                          | 23              |
| 9          | Second Level DFD – User Management                 | 24              |
| 10         | Second Level DFD-Internship Processing Unit        | 25              |
| 11         | Second Level DFD-Recommendation and Scam Detection | 25              |

LIST OF TABLES

| NO. | NAME                   | PAGE NO. |
|-----|------------------------|----------|
| 1   | Student Entity         | 28       |
| 2   | Company Entity         | 29       |
| 3   | Application and Report | 29       |
| 4   | Internship Posting     | 30       |

# **1 INTRODUCTION**

---

- **INTRODUCTION**
- **AIMS AND OBJECTIVE OF THE WORK**
- **BRIEF LITERATURE REVIEW**
- **PROBLEM DEFINITION**
- **PLAN OF THEIR WORK**



# 1.Introduction

## 1.1 Introduction

Internships play a vital role in bridging the gap between academic learning and industry requirements. However, in today's competitive environment, students often face challenges in identifying genuine internship opportunities and improving their employability skills. With the growing number of online internship postings, the risk of fraudulent or misleading listings has also increased, which can lead to financial loss and wasted effort for students.

To address these challenges, SecureSmart Intern is proposed as an AI-powered web platform that assists students in discovering suitable internships while ensuring safety and reliability. The platform enables students to create detailed skill-based profiles, explore tailored internship opportunities, track applications, and receive guidance on missing skills. It also incorporates a fraud detection mechanism to flag potentially fake internship postings, allowing students to make informed decisions.

## 1.2 Aims and Objective of the Work

The main aim of the project is to develop a secure and intelligent platform for internship recommendation and fraud detection. The specific objectives include:

1. To enable students to create detailed profiles including personal details, skills, and educational qualifications.
2. To provide AI-assisted internship recommendations based on skill matching and gaps.
3. To detect and flag potentially fake or suspicious internship postings using keyword and heuristic analysis.
4. To provide companies with a platform to post internships and manage applications efficiently.
5. To improve student employability by highlighting missing skills and providing actionable guidance.

## 1.3 Brief Literature Review

Several platforms currently exist for internship discovery, such as Internshala, LinkedIn, and Indeed. These platforms primarily rely on keyword-based search and manual verification processes. While they provide large databases of internships, they do not actively analyse the skill gap of students or assess the authenticity of postings in real-time.

Research in AI-assisted recommendation systems has shown that semantic analysis and skill matching significantly improve the relevance of suggested opportunities. Similarly, studies on fraud detection indicate that heuristic-based analysis of postings, combined with community feedback, can effectively identify scams. This project integrates these techniques to provide a combined solution of personalized recommendations and scam detection, which is not fully addressed in existing platforms.

## 1.4 Problem Definition

Students often struggle to:

- Identify internships that match their skills and career goals.
- Evaluate the credibility of internship postings, leading to potential scams.
- Track applications and manage multiple internships effectively.

At the same time, companies need a reliable platform to reach skilled candidates while avoiding spam or irrelevant applications. Existing platforms fail to address these challenges comprehensively.

Problem Statement:

“To design a web-based system that provides AI-powered internship recommendations while proactively detecting and flagging fraudulent postings, ensuring a safe and efficient internship search experience for students and a reliable recruitment process for companies.”

## 1.5 Plan of Their Work

The project development is divided into several stages:

1. Requirement Analysis: Understanding the needs of students and companies, and identifying key functional and non-functional requirements.
2. System Design: Creating all necessary diagrams including Use Case, ER, Class, DFD, Sequence, and Activity diagrams to model the system comprehensively.
3. Database Design: Translating the ER diagram into database schemas, including tables for students, companies, internships, applications, and skills.
4. Frontend Development: Designing student and company dashboards, registration forms, profile management pages, internship listing pages, and application interfaces using a web framework.
5. Backend Development: Implementing APIs for authentication, internship management, application submission, and scam detection logic.
6. Integration and Testing: Connecting frontend with backend, testing all functionalities including skill-based recommendations, application tracking, and risk scoring.
7. Deployment and Documentation: Preparing final deliverables including system screenshots, report, and demonstration of working features.

This structured approach ensures parallel development by team members and allows incremental testing, improving efficiency and reducing errors during implementation.

The flowchart illustrates the complete development lifecycle of the SecureSmart Intern project. It starts with requirement analysis and problem identification, followed by literature review and system design. The architecture and database are designed before proceeding with frontend and backend development. Machine learning models for internship recommendation and scam detection are developed and integrated into the system. The project undergoes testing and validation, and any identified errors are corrected through optimization and re-testing. After successful validation, the system is deployed, results are analyzed, and final documentation is prepared.

### Flowchart of Project Development Plan

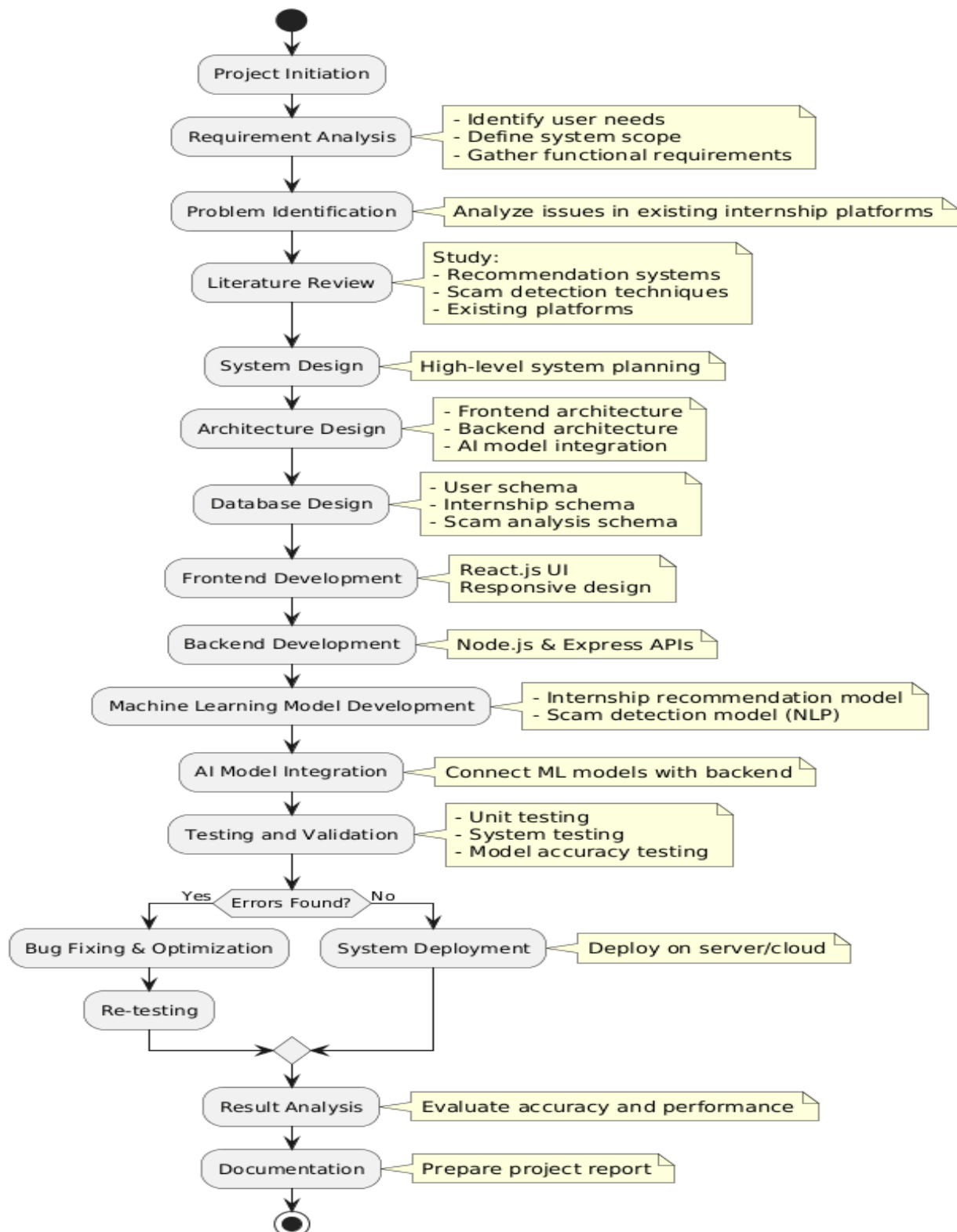


FIG 1: Flowchart of Project Development Plan

## **2 TECHNOLOGY AND LITERATURE REVIEW**

---

- **TECHNOLOGY**
- **LITERATURE REVIEW**
- **RESEARCH GAP AND MOTIVATION**

## 2.1 Technology Overview

Modern web-based platforms are commonly developed using a multi-tier architecture consisting of a presentation layer, application layer, and data layer. The proposed system follows this approach to ensure scalability, security, and maintainability.

The **presentation layer** provides interactive user interfaces for students and companies, enabling profile management, internship browsing, and application tracking. The **application layer** handles business logic such as authentication, internship recommendations, application processing, and scam risk evaluation. The **data layer** is responsible for storing user profiles, internship details, skill information, application records, and reports in a structured manner.

To enhance system intelligence, **AI-assisted techniques** such as skill similarity analysis and heuristic-based fraud detection are employed. Skill matching involves comparing student skills with internship requirements to generate relevance scores. Scam detection uses predefined rules and keyword analysis to identify suspicious patterns such as unrealistic stipends, urgent payment requests, or incomplete company information. These techniques improve the accuracy of recommendations and enhance platform safety.

## 2.2 Literature Review

Existing internship and job search platforms primarily rely on keyword-based search and manual verification processes. While these systems provide a wide range of opportunities, they often lack personalized guidance and real-time assessment of internship authenticity. Students are required to independently evaluate postings, which increases the risk of falling victim to fraudulent offers.

Research in **recommendation systems** highlights the effectiveness of skill-based and semantic matching techniques over traditional keyword matching. Studies indicate that analyzing user skills, preferences, and interaction behavior leads to more relevant and personalized recommendations. Similarly, research on **online fraud detection** emphasizes the use of linguistic pattern analysis, anomaly detection, and user feedback mechanisms to identify suspicious content.

Recent studies suggest that combining **rule-based analysis with user-generated reports** improves fraud detection accuracy and adaptability to evolving scam patterns. These findings support the design approach adopted in SecureSmart Intern, where internship recommendations and scam detection are integrated into a single platform to ensure both usability and security.

### 2.3 Research Gap and Motivation

Although several platforms offer internship listings, very few systems integrate **personalized skill-based recommendations** with **proactive scam detection**. Existing solutions either focus on opportunity discovery or rely heavily on manual reporting after fraud has occurred. There is a clear gap in providing a unified system that assists students in both **career development** and **online safety**.

The proposed system aims to bridge this gap by combining AI-assisted internship recommendations with an automated scam detection mechanism. This approach not only improves the relevance of opportunities presented to students but also creates a safer digital environment for internship exploration.

### **3 SYSTEM REQUIREMENT STUDY**

---

- **USER CHARACTERISTICS**
- **HARDWARE AND SOFTWARE REQUIREMENTS**
- **ASSUMPTIONS AND DEPENDENCIES**

### 3. System Requirements Study

This chapter describes the system requirements necessary for the development and deployment of the proposed system, **SecureSmart Intern**. It focuses on identifying the characteristics of users interacting with the system, the hardware and software resources required, and the assumptions and dependencies considered during development. A clear understanding of these requirements ensures that the system is reliable, scalable, and capable of fulfilling its objectives effectively.

#### 3.1 User Characteristics

The SecureSmart Intern platform is designed for multiple types of users, each having distinct roles and access privileges within the system.

##### 3.1.1 Student Users

Students are the primary users of the system who seek internship opportunities.

###### Characteristics:

- Undergraduate students from various academic backgrounds
- Basic knowledge of using web-based applications
- Access the system through desktops, laptops, or mobile devices
- Varying levels of technical and professional skills

###### Activities:

- Register and manage personal profiles
- Add educational details, skills, and resumes
- View AI-based internship recommendations
- Apply for internships and track application status

##### 3.1.2 Company Users

Company users represent organizations offering internships.

###### Characteristics:

- HR personnel or company representatives
- Familiar with online recruitment platforms
- Require secure access to manage postings and applicants

###### Activities:

- Create and update company profiles
- Post internship opportunities
- Review student applications
- Shortlist or reject candidates



### 3.1.3 Administrator

The administrator is responsible for monitoring and managing the overall system.

**Characteristics:**

- Authorized personnel with complete system access
- Technical knowledge of system operations

**Activities:**

- Verify company registrations
- Monitor internship postings
- Handle reported or suspicious activities
- Ensure data security and system maintenance

## 3.2 Hardware and Software Requirements

This section specifies the minimum hardware and software requirements needed to run the SecureSmart Intern platform efficiently.

### 3.2.1 Hardware Requirements

**For End Users (Students and Companies):**

- Processor: Dual-core processor or higher
- RAM: Minimum 4 GB
- Storage: Minimum 500 MB free disk space
- Device: Desktop, Laptop, Tablet, or Smartphone
- Internet Connection: Stable broadband or mobile internet

**For Server:**

- Processor: Quad-core processor
- RAM: Minimum 8 GB
- Storage: SSD with at least 50 GB free space
- Network: High-speed internet connectivity

### 3.2.2 Software Requirements

**Client-Side Software:**

- Operating System: Windows, macOS, Linux, Android, or iOS
- Web Browser:
  - Google Chrome
  - Mozilla Firefox
  - Microsoft Edge

**Server-Side Software:**

- Operating System: Linux (Ubuntu) or Windows Server
- Frontend Technologies: HTML, CSS, JavaScript, React.js
- Backend Technologies: Node.js or Python (Flask/Django)
- Database: PostgreSQL or MySQL
- Authentication: JSON Web Tokens (JWT)
- Development Tools: Visual Studio Code, Git

**3.3 Assumptions and Dependencies****3.3.1 Assumptions**

The following assumptions are made during system development:

- Users have access to a reliable internet connection
- Students provide accurate and updated profile information
- Companies post genuine internship opportunities
- The system initially supports a moderate number of users
- Users possess basic knowledge of web applications

**3.3.2 Dependencies**

The system performance depends on:

- Availability of internet connectivity
- Proper functioning of server and database services
- Reliability of third-party libraries and frameworks
- Secure authentication and authorization mechanisms
- Regular system updates and maintenance

## **4 SYSTEM DESIGN**

---

- **CLASS DIAGRAM**
- **USE CASE DIAGRAM**
- **ENTITY RELATIONSHIP DIAGRAM**
- **SEQUENCE DIAGRAM**
- **ACTIVITY DIAGRAM**
- **DATA FLOW DIAGRAM**

## 4.1 Class Diagram

The Class Diagram represents the static structure of the SecureSmart Intern system. It defines the major classes, their attributes, methods, and relationships. The diagram shows a generalized User class, which is inherited by Student and Company, enabling role-based functionality.

The student class manages profile details, skills, internship applications, and reporting of suspicious internships. The Company class handles internship postings, application review, and communication with students. The Internship class stores internship-related information including risk score and posting status. The Application class acts as an associative entity connecting students and internships. Supporting classes such as Skill, Chat Message, Recommendation Engine, Scam Detection, and Report provide AI-based recommendations, scam analysis, communication, and safety reporting. This diagram serves as the foundation for object-oriented implementation.

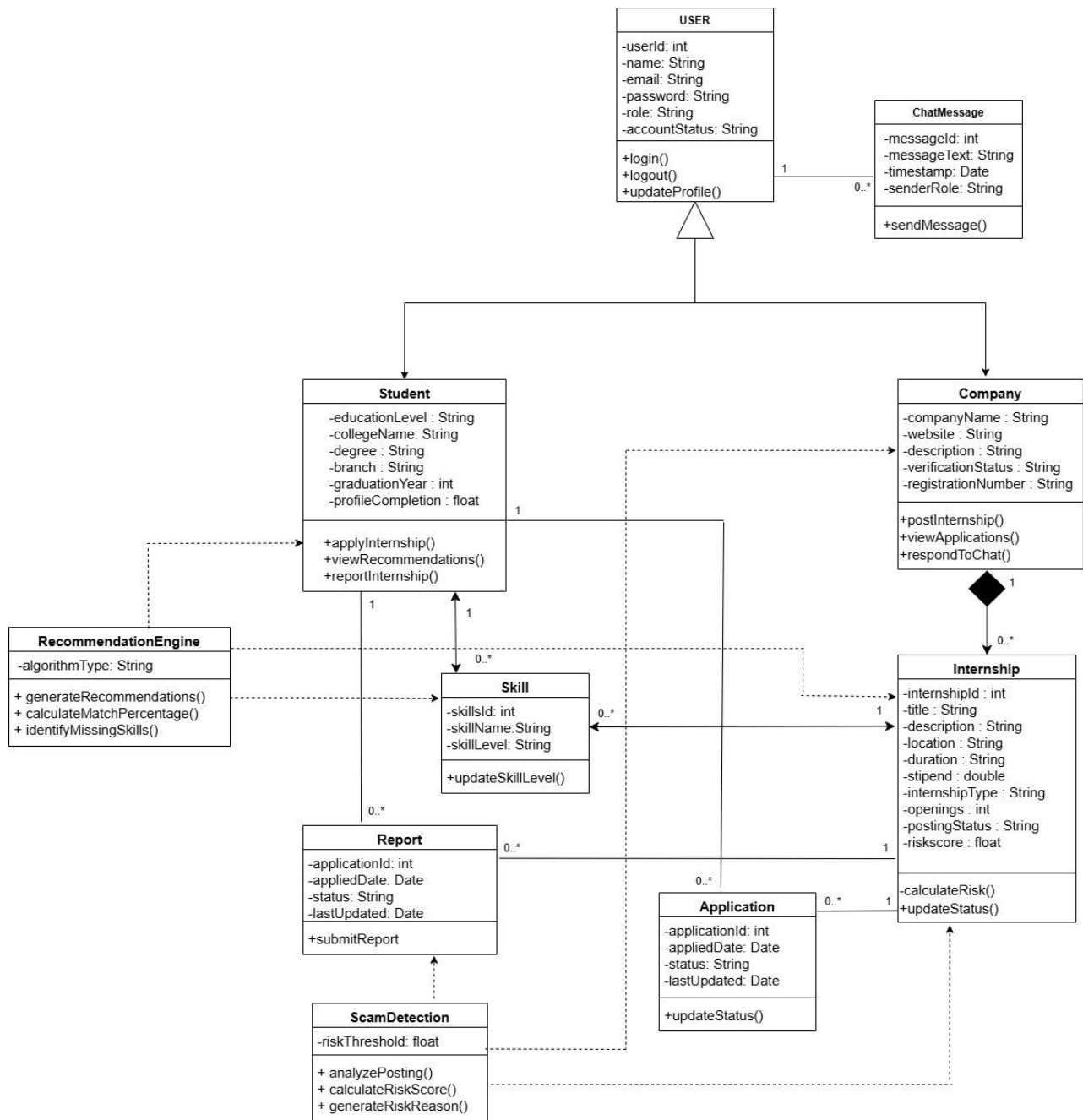


FIG 2: Class Diagram

## 4.2 Use Case Diagram

The Use Case Diagram illustrates the functional behaviour of the system from the perspective of three actors: Student, Company, and Admin. It defines how each actor interacts with the system and the services available to them.

Students can register/login, manage profiles, view recommended internships, search and apply for internships, track application status, report suspicious internships, and receive scam alerts. Companies can post internships, manage listings, view applications, communicate with students, and shortlist or reject candidates. The Admin actor oversees platform activities by reviewing flagged internships, user reports, and taking moderation actions. This diagram provides a high-level view of system functionality and user responsibilities.



FIG 3: Use Case Diagram

### 4.3 Entity Relationship Diagram (ER Diagram)

The Entity Relationship Diagram represents the logical database design of the system. It identifies key entities such as User, Student, Company, Internship, Skill, Application, ChatMessage, and Report, along with their attributes and relationships.

The diagram uses an ISA relationship to represent inheritance between User, Student, and Company. Associative entities such as StudentSkill and InternshipSkill manage many-to-many relationships between students and skills, and internships and required skills. The Application entity connects students with internships, while Report captures user-submitted complaints related to internships or companies. This ER diagram ensures data consistency and supports efficient database design.

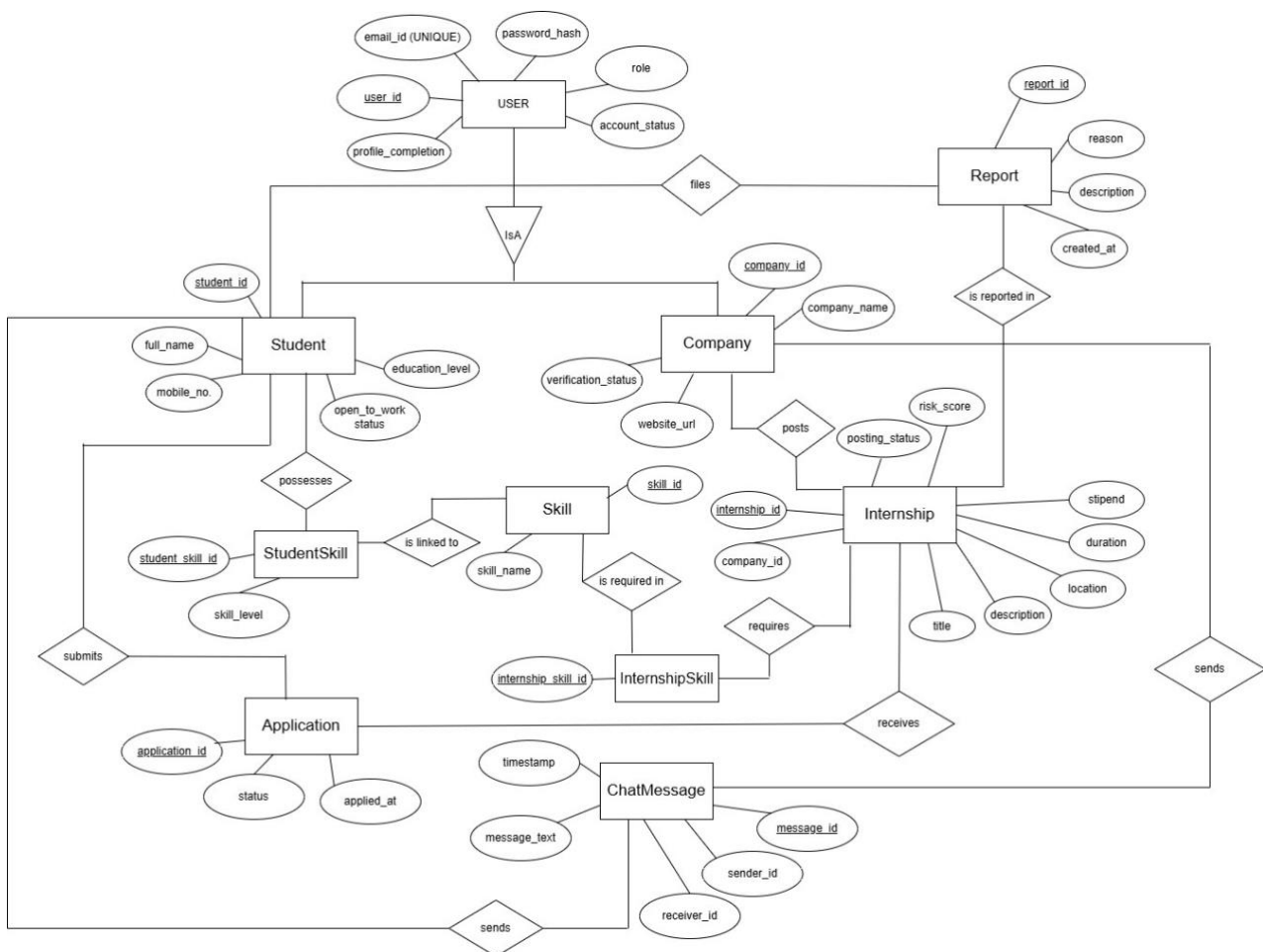


FIG 4: Entity Relationship diagram



## 4.4 Sequence Diagram

The Sequence Diagram illustrates the dynamic interaction between system components over time during the internship recommendation and application process. It shows the communication flow among the Student, Web Interface, AI Recommendation Engine, Scam Detection Module, Database, Company, and Admin.

The sequence begins with student login and authentication, followed by profile updates and skill submission. Internship recommendations are generated by the AI engine, and postings are analysed by the scam detection module to produce risk scores and warning messages. The student can then apply for internships, after which application status updates are managed and notifications are sent. Admin interactions include monitoring flagged internships and reports. This diagram clearly depicts message flow and system coordination.

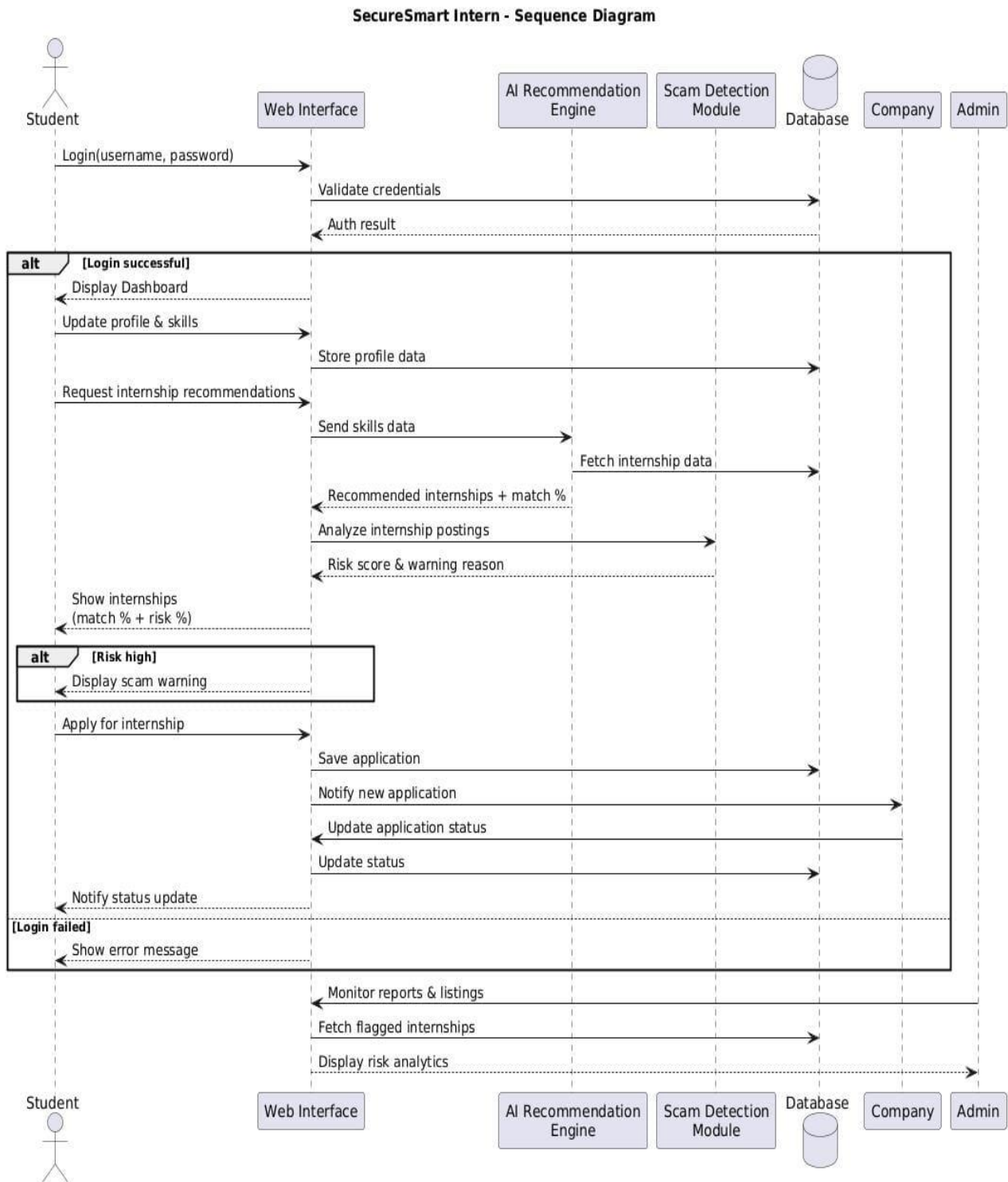


FIG 5: Sequence Diagram

## 4.5 Activity Diagram

The Activity Diagram represents the workflow of student actions within the system. It begins with user authentication and proceeds through profile management, skill updates, internship search, and recommendation generation.

The diagram includes decision points such as AI match availability and risk score evaluation. Based on scam detection results, internships are marked as safe or flagged with warnings. The workflow continues through application submission, status updates, and student notification. This diagram helps visualize control flow, decision-making, and parallel activities within the system.

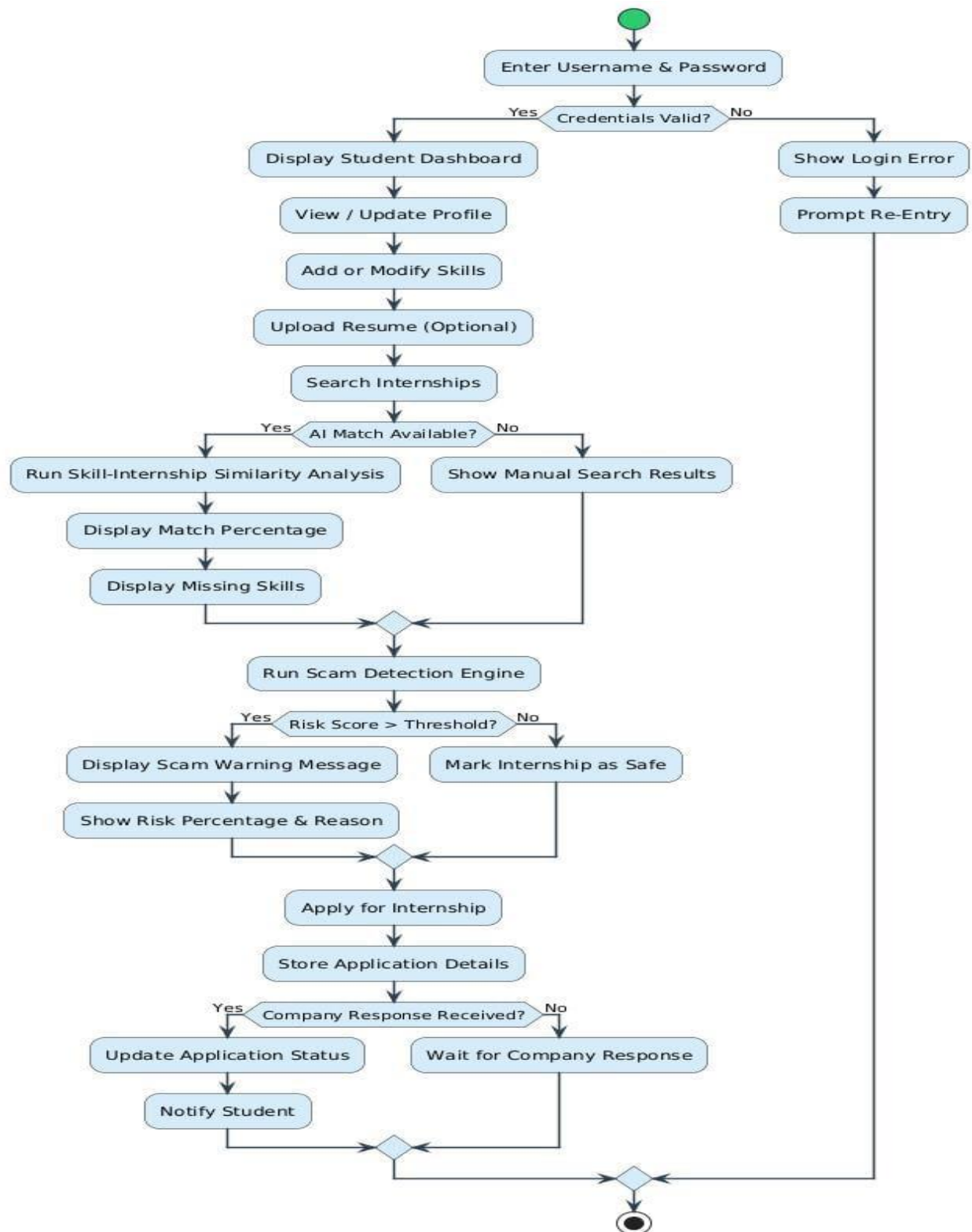


FIG 6: Activity Diagram

## 4.6 Data Flow Diagram

The Data Flow Diagram of the SecureSmart Intern system represents the flow of data between students, companies, and the system for internship recommendation and scam detection. Overall, the DFD illustrates how data flows securely and systematically to support personalized recommendations, transparent application tracking, and proactive scam prevention within the platform.

**4.6.1 Context Level DFD (Level 0):** At Level-0, the system is viewed as a single process where students provide profile details and internship preferences, companies submit internship postings, and the system returns recommendations, application statuses, and risk alerts while enabling reporting of suspicious activities.

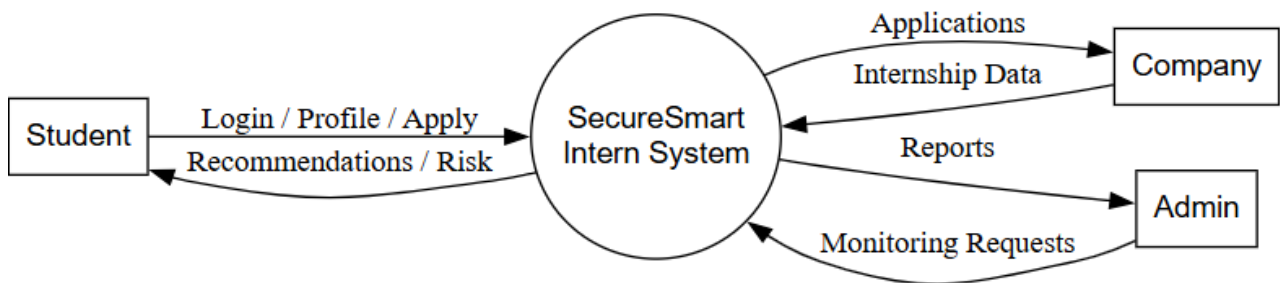


FIG 7.1: Context-Level DFD

**4.6.2 First Level DFD (Level 1):** Level-1 decomposes the system into major processes such as user authentication, profile management, internship posting, recommendation generation, application management, chat communication, and scam detection, each interacting with respective data stores like student profiles, company details, internship records, applications, skills, and reports

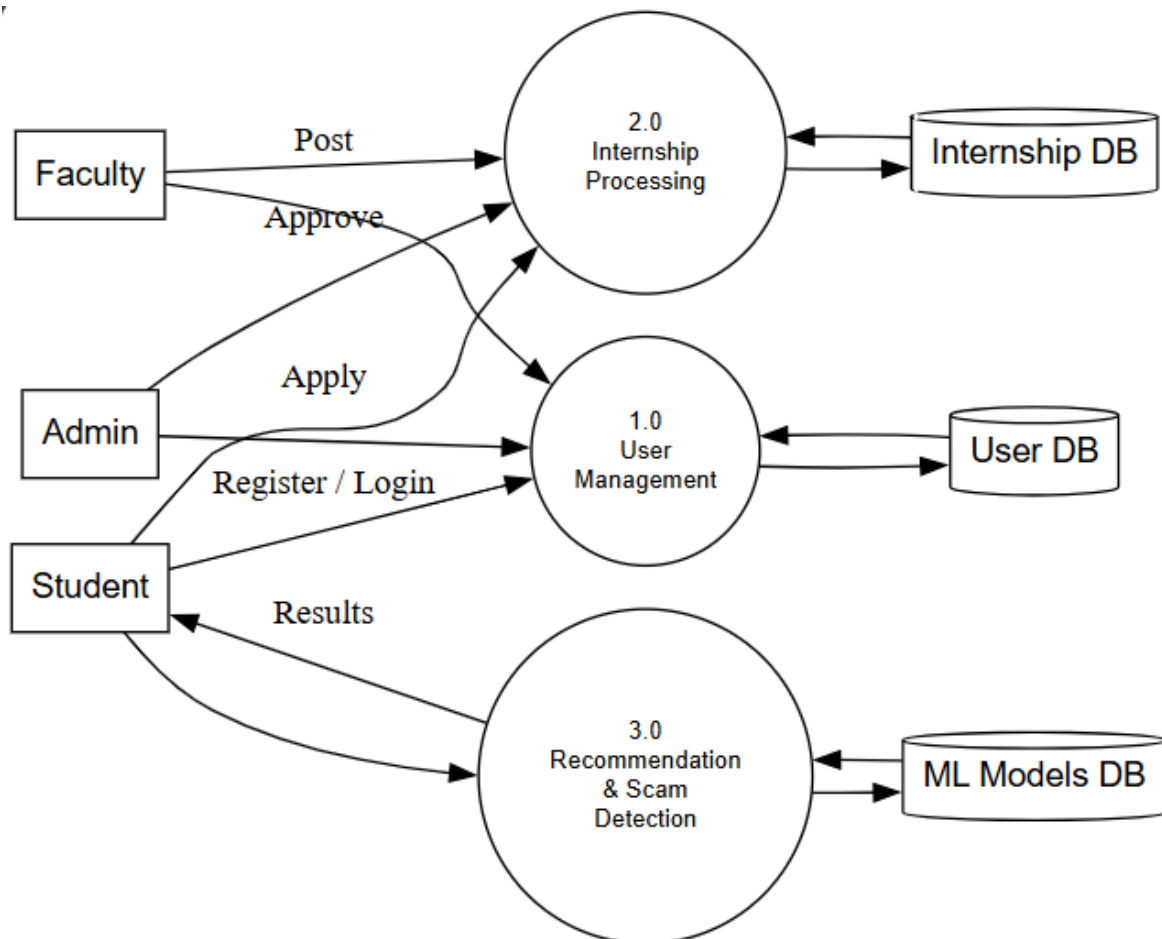


FIG 7.2: First-Level DFD

**4.6.3 Second Level DFD (Level 2):** At Level-2, detailed processing is shown where student skills are analyzed to calculate match percentages, internship postings are evaluated using heuristic rules to generate risk scores and reasons, user reports are stored and used to update risk levels, and application statuses are updated based on company actions.

- **User Management (Level 2)**

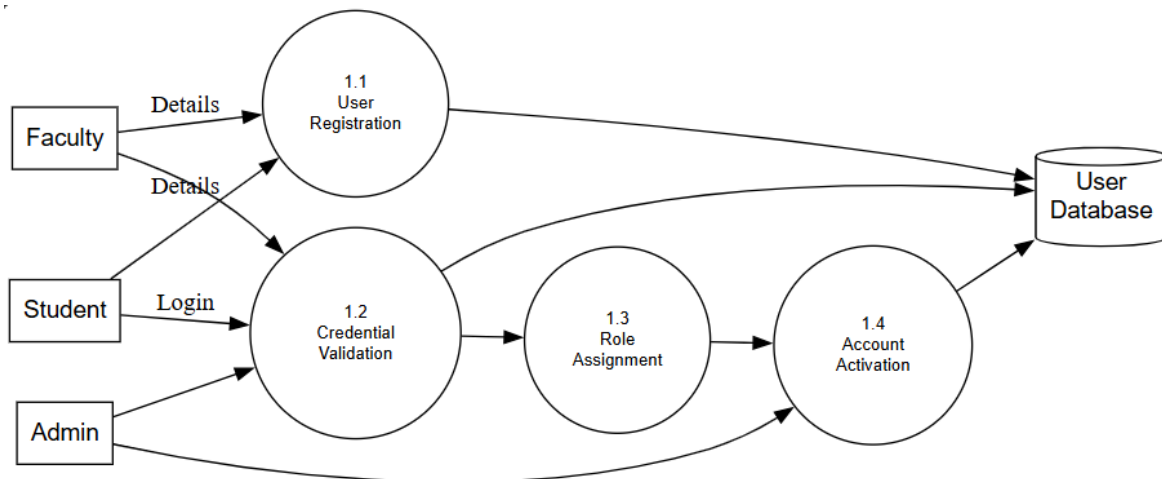


FIG 7.3.1: Second-Level DFD-User Management

- **Internship Processing Unit (Level 2)**

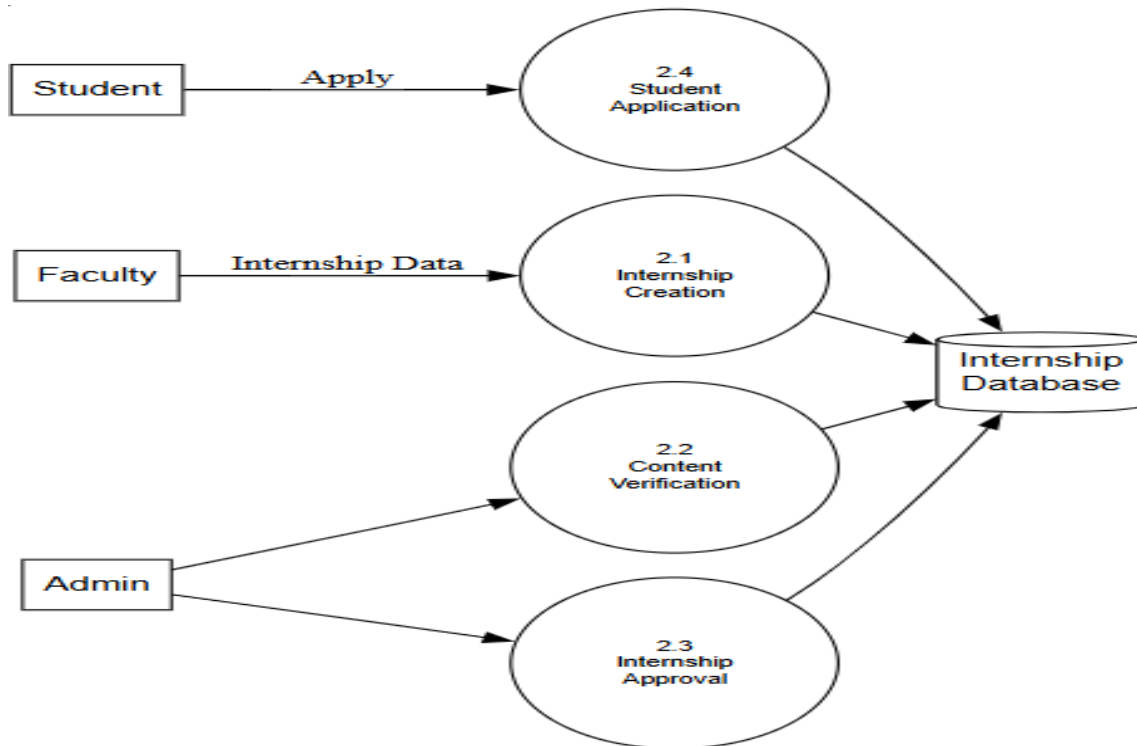


FIG 7.3.2: Second-Level DFD-Internship Processing Unit

- **Recommendation and Scam Detection (Level 2)**

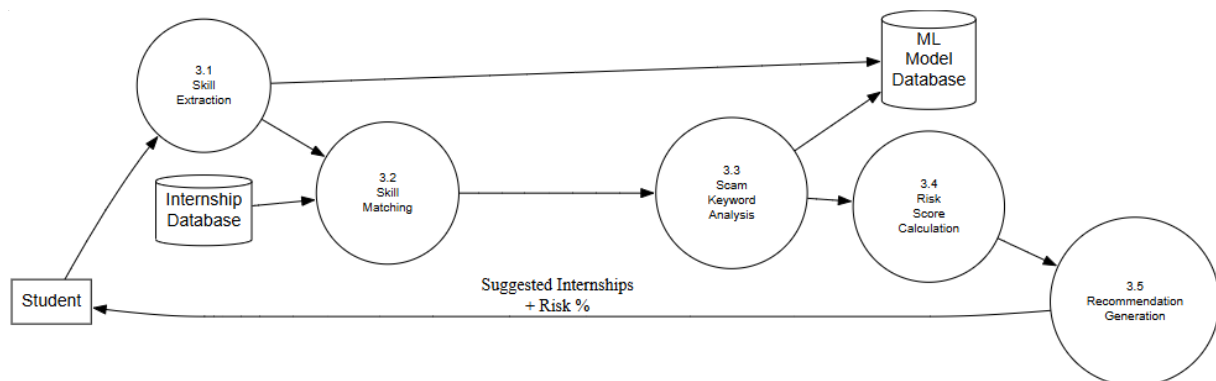


FIG 7.3.3: Second-Level DFD-Recommendation and Scam Detection



## **5 DATA DICTIONARY**

---

- **STUDENT ENTITY**
- **COMPANY ENTITY**
- **APPLICATION AND REPORT**
- **INTERNSHIP POSTING**

## 5. Data Dictionary

The data dictionary describes the database structure used in the SecureSmart Intern system. It defines the key entities, their attributes, data types, and constraints required to support system functionalities such as user registration, internship posting, application tracking, skill matching, and scam detection.

The database includes tables for managing student profiles, company details, internship postings, and internship applications. Unique identifiers and primary keys are used to ensure data integrity, while foreign keys establish relationships between students, companies, internships, and applications. Sensitive information such as passwords is securely stored in hashed format, and optional fields are provided for profile customization and document uploads.

AI-generated attributes such as skill match percentage and risk score are included to support internship recommendation and scam detection features. Enumeration fields are used to manage application status, verification status, and reporting reasons, ensuring consistency in system operations.

Overall, the data dictionary provides a clear and structured reference for database implementation, helping maintain consistency, scalability, and reliability of the SecureSmart Intern platform.

## 5.1 Student Entity

| Student_id    | Integer      | Primary Key      | Unique identifier for each student.        |
|---------------|--------------|------------------|--------------------------------------------|
| full_name     | Varchar(100) | Not Null         | Student's full name.                       |
| email         | Varchar(100) | Unique; Not Null | Official login email.                      |
| phone_no      | Varchar(15)  | Optional         | Contact number.                            |
| password      | Varchar(255) | Not Null         | Hashed password for security.              |
| password      | Varchar(255) | Not Null         | Hashed password for security.              |
| college_name  | Varchar(150) | Not Null         | Name of the institution.                   |
| passing_year  | Year         | Not Null         | Graduation year.                           |
| skills_list   | Text/Array   | Not Null         | List of technical/soft skills.             |
| cv_resume     | Blob/Link    | Optional         | URL or binary file of the uploaded resume. |
| description   | Text         | Optional         | Short "About Me" bio.                      |
| profile_photo | Link         | Optional         | URL to the stored image.                   |
| work_status   | Enum         | Default: 'Open'  | Options: 'Open to work'; 'Hired'.          |

Table 1: Student Entity Table

5.2 Company Entity

| Field Name     | Data Type    | Constraint         | Description                                 |
|----------------|--------------|--------------------|---------------------------------------------|
| company_id     | Integer      | Primary Key        | Unique identifier for each company.         |
| company_name   | Varchar(100) | Not Null           | Registered legal name.                      |
| official_email | Varchar(100) | Unique; Not Null   | Primary contact email.                      |
| website_url    | Varchar(255) | Not Null           | Link to the company website.                |
| reg_number     | Varchar(50)  | Optional           | Official business registration ID.          |
| verify_status  | Enum         | Default: 'Pending' | Options: 'Pending'; 'Verified'; 'Rejected'. |

Table 2: Company Entity Table

5.3 Application and Report

| Field Name    | Data Type       | Constraint         | Description                                                 |
|---------------|-----------------|--------------------|-------------------------------------------------------------|
| app_id        | Integer         | Primary Key        | Unique application record.                                  |
| student_id    | Integer         | Foreign Key        | Links to the Student.                                       |
| job_id        | Integer         | Foreign Key        | Links to the Internship.                                    |
| status        | Enum            | Default: 'Applied' | Options: 'Applied'; 'Under Review'; 'Accepted'; 'Rejected'. |
| match_percent | Integer (0-100) | System Gen.        | AI-calculated skill similarity score.                       |
| report_reason | Enum            | Optional           | Reasons: 'Payment request'; 'Fake Info'; 'Spam'.            |

Table 3: Application and Report

5.4 Internship Posting

| Field Name      | Data Type       | Constraint  | Description                           |
|-----------------|-----------------|-------------|---------------------------------------|
| job_id          | Integer         | Primary Key | Unique identifier for the internship. |
| company_id      | Integer         | Foreign Key | Links to the Company Entity.          |
| title           | Varchar(100)    | Not Null    | Role title (e.g. "Web Dev Intern").   |
| location        | Varchar(100)    | Not Null    | Remote or City name.                  |
| duration        | Varchar(50)     | Not Null    | e.g. "3 Months".                      |
| required_skills | Text/Array      | Not Null    | Skills needed for the role.           |
| stipend_amount  | Decimal         | Not Null    | Monthly payment value.                |
| risk_score      | Integer (0-100) | Computed    | Calculated by AI based on heuristics. |
| risk_reason     | Text            | System Gen. | Explanation for the risk flag.        |

Table 4: Internship and Posting

## **6 SNAPSHOTS**

---

- **WEBSITE HOMEPAGE**
- **STUDENT**
- **COMPANY**

6.1 Website Homepage

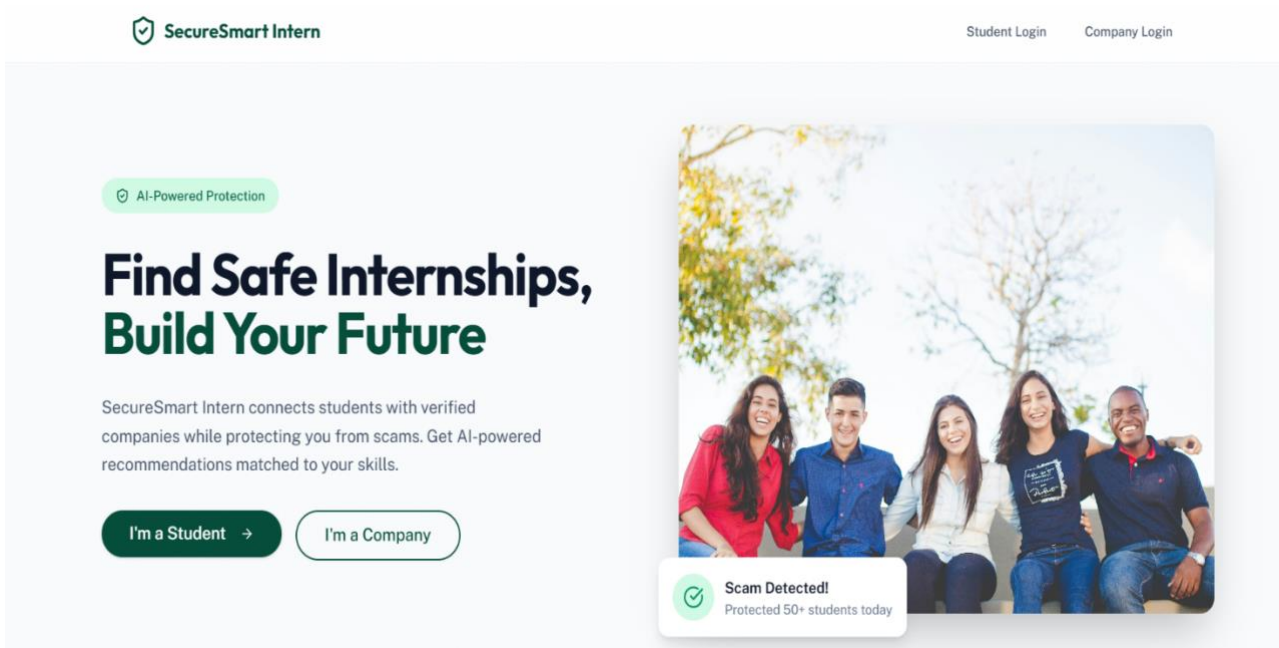


FIG 8.1: Website Homepage

6.2 Student

6.2.1 Student Registration:

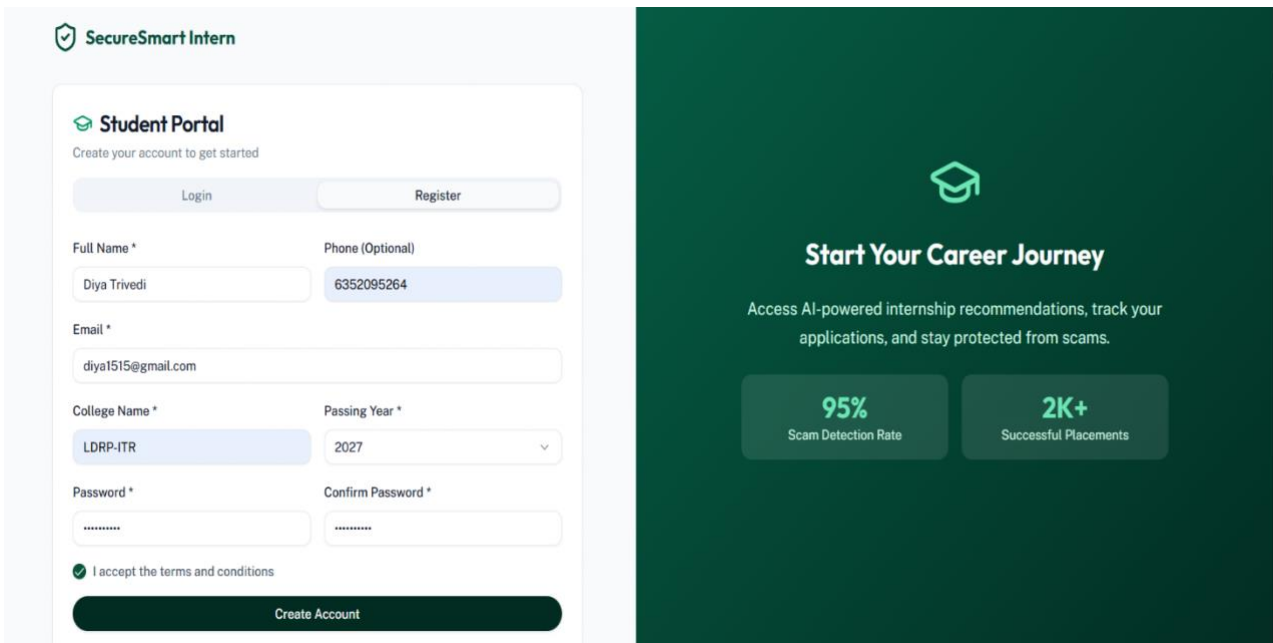


FIG 8.2: Student Registration

6.2.2 Student Login:

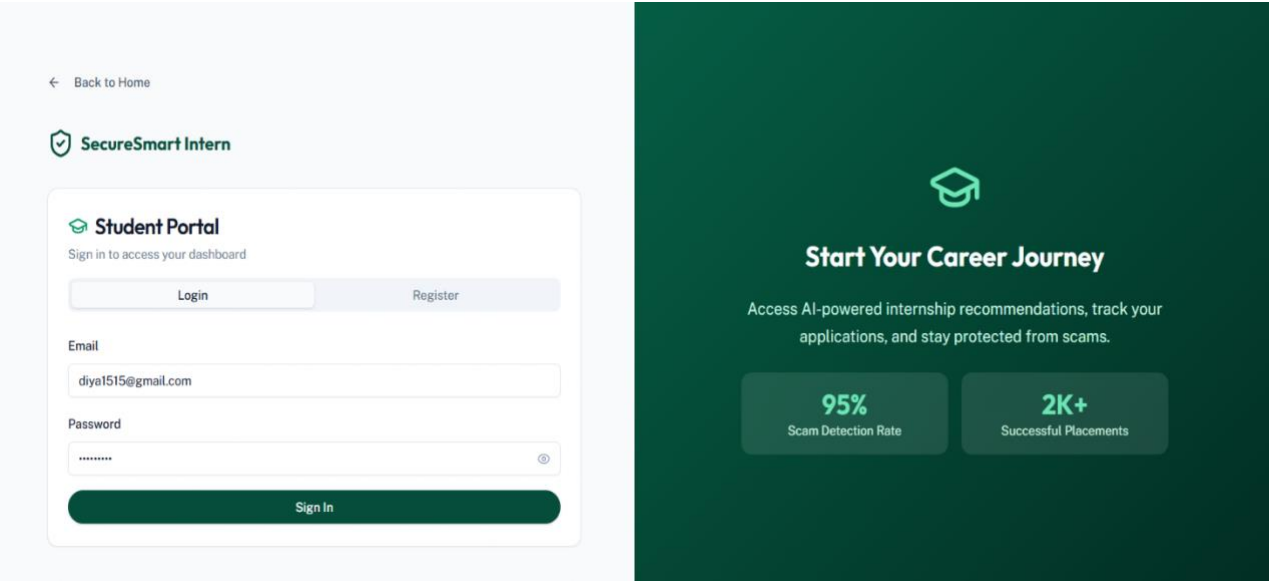


FIG 8.3: Student Login

6.2.3 Student Homepage:

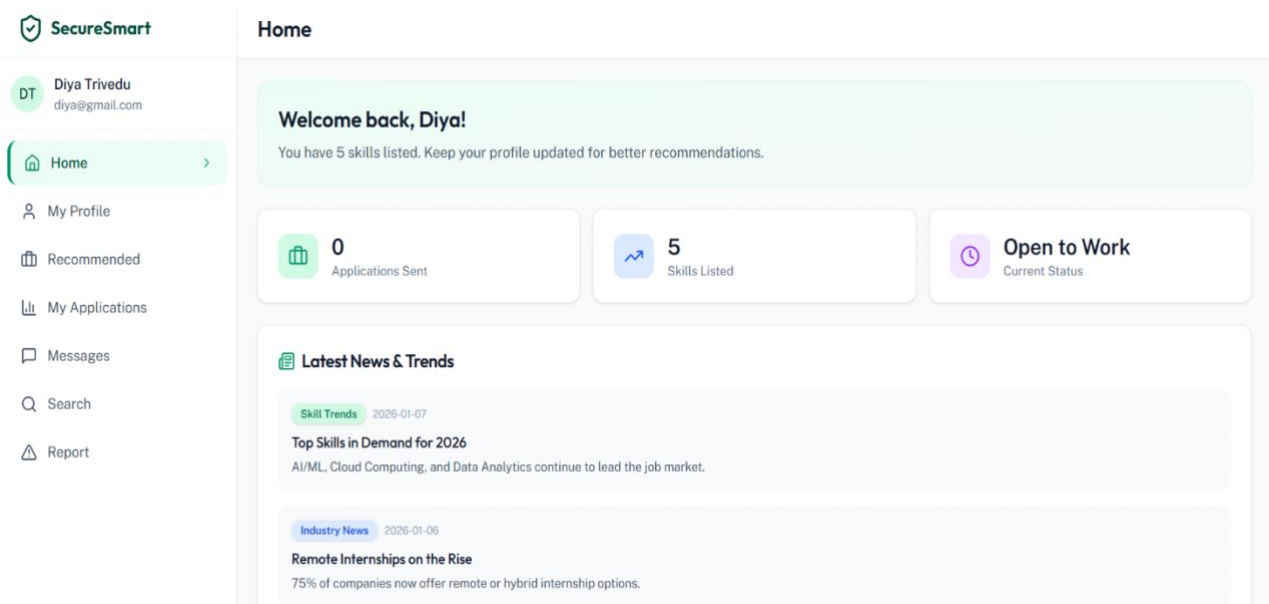


FIG 8.4: Student Homepage



6.2.4 Student Profile:

DT

Diya Trivedi

diya15@gmail.com

Home

**My Profile**

Recommended

My Applications

Messages

Search

Report

My Profile

Personal Information

Full Name

Diya Trivedi

Phone Number

6352095264

College Name

LDRP-ITR

Passing Year

2027

Status

Open to Work

Bio / Description

Tell us about yourself...

Save Changes

Skills

Add a skill (e.g., Python, React, Data Analysis)

+

Python

AI

ML

REACT

Resume

No resume uploaded yet.

Upload Resume

Profile Photo

Upload Photo

Email (read-only)

diya15@gmail.com

FIG 8.5: Student Profile

6.2.5 Recommendation:

DT

Diya Trivedi

diya15@gmail.com

Home

My Profile

**Recommended**

My Applications

Messages

Search

Report

Recommended

Found 4 internships matching your profile

Software Development Intern

Test Company 063403

Join our team as a software development intern. You'll work on real projects and gain valuable experience.

San Francisco, CA

3 months

\$2000/month

Remote

Required Skills:

Python

JavaScript (missing)

React

66% Match

Low Risk

25%

Very brief, generic description with no specifics about projects, responsibilities, team, or tech stack beyond a few skills

Missing key company details (company name, website, contact email/domain, hiring manager, interview process)

Location listed as San Francisco, CA while Type is Remote (not necessarily fraudulent, but slightly inconsistent without clarification)

Apply Now

Software Development Intern

66% Match

Low Risk 25%

FIG 8.6: Recommendations

LDRP-ITR | CE Department

34

6.2.6 Report:

 SecureSmart

DT

Diya Trivedi

diya15@gmail.com

Home

My Profile

Recommended

My Applications

Messages

Search

Report

Report

 **Report Internship / Company**

Help us maintain a safe platform by reporting suspicious internships or companies.

Select Internship \*

Software Development Intern - Test Company 063257

Company: Test Company 063257

Internship: Software Development Intern

Report Reason \*

Fake company details

Additional Details (Optional)

This company is impersonating to be the original company already existing and has misleading information and fake company details which are identified as they differ slightly from original company details.

Note: Submitting false reports may result in action against your account. Please only report genuine concerns.

Submit Report

FIG 8.7: Report

6.3 Company:  
6.3.1 Company Registration:

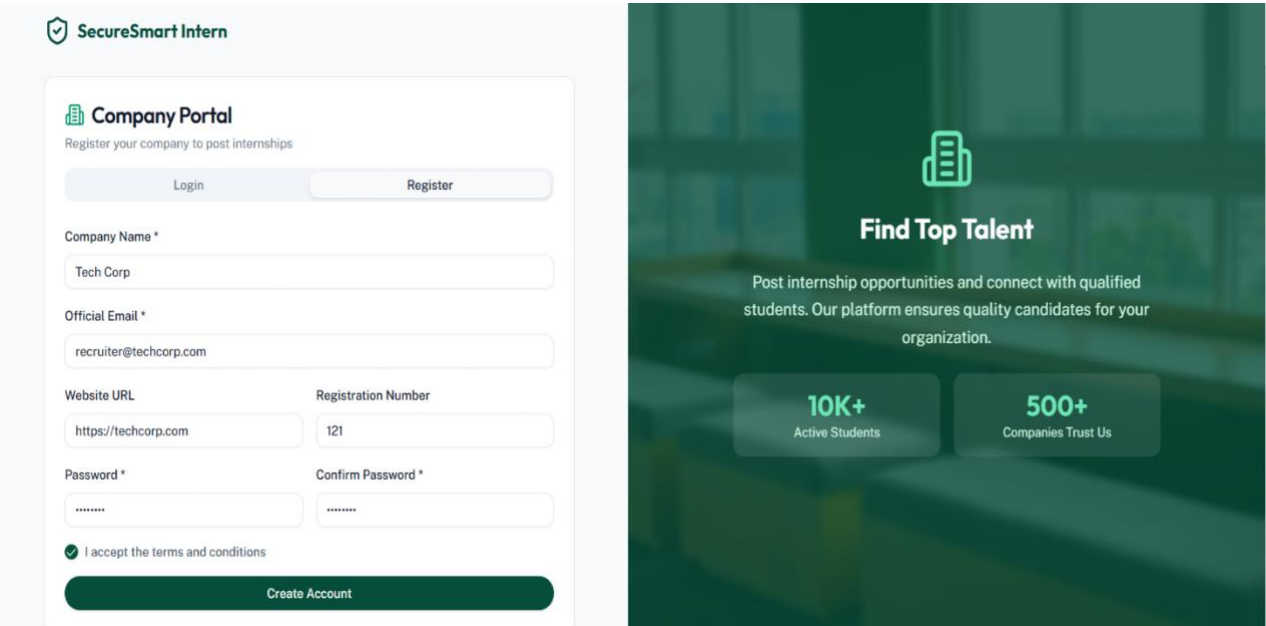


FIG 8.8 Company Registration

6.3.2 Company Login:

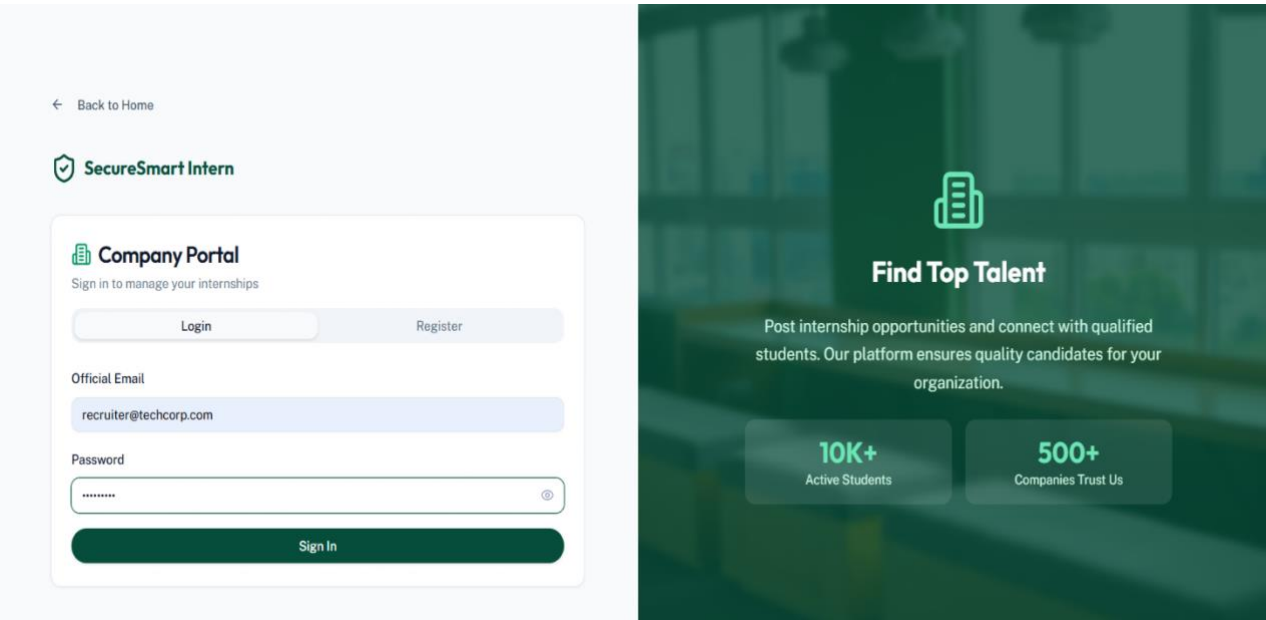


FIG 8.9 Company Login

6.3.3 Company Homepage:

SecureSmart

TC Tech Corp  
recruiter@techcorp.com

Home

Company Profile

Post Internship

Applications

Messages

Search Students

Notifications

Home

Company Information

Company Name  
Tech Corp

Website URL  
https://techcorp.com

Company Description  
Tech Corp Innovations is a fast-growing technology company founded in 2021, working in the areas of web development, data analytics, and AI solutions. Our mission is to bridge the gap between academic learning and industry requirements by providing practical exposure to students and fresh graduates. We work on real-world projects for startups and small businesses, offering innovative and scalable solutions.

Save Changes

Verification Status

Pending Verification

Complete your profile to get verified and build trust with students.

Official Email (read-only)  
recruiter@techcorp.com

Registration Number  
121

Documents

No documents uploaded yet.

FIG 8.10: Company Homepage

6.3.4 Company Post Internships:

SecureSmart

T Tatvasoft  
tatva@gmail.com

Home

Company Profile

Post Internship

Applications

Messages

Search Students

Notifications

Home

Post New Internship

Fill in the details below. Our AI will analyze your posting for potential issues.

Internship Title \*  
e.g., Software Development Intern

Description \*  
Describe the role, responsibilities, and what interns will learn...

Required Skills \*  
Add a skill (e.g., Python, React) Add

Location \*  
e.g., New York, NY or Remote

Duration \*  
e.g., 3 months

Internship Type \*  
Remote

Stipend \*  
e.g., \$1000/month

Number of Openings  
1

AI Scam Analysis: Your posting will be automatically analyzed for potential red flags. This helps maintain trust and protects both students and companies.

Post Internship

Made with Emergent

FIG 8.11: Company Post Internships

LDRP-ITR | CE Department

37

## **7 RESULT, DISCUSSION AND CONCLUSION**

---

- **RESULT**
- **DISCUSSION**
- **CONCLUSION**

### 7.1 Result

The SecureSmart Intern system successfully fulfills its intended objectives of providing personalized internship recommendations and detecting potentially fraudulent internship postings. The platform allows students to create detailed profiles, add skills, and receive AI-assisted internship suggestions based on skill matching and relevance scores. Students can apply for internships, track application status, and receive alerts related to suspicious postings.

Companies are able to post internship opportunities, manage applications, and shortlist candidates efficiently through a dedicated dashboard. The scam detection module evaluates internship listings using predefined heuristics and assigns a risk score, enabling early identification of suspicious or fake postings. Overall, the system demonstrates reliable performance, user-friendly interaction, and secure handling of user data.

### 7.2 Discussion

The implementation of SecureSmart Intern highlights the effectiveness of integrating AI-based recommendation techniques with security-focused scam detection mechanisms. Skill-based matching improves the relevance of internship recommendations, reducing the time and effort required by students to find suitable opportunities. The scam detection component enhances trust and safety by alerting users about potentially harmful listings.

While the system performs well for moderate user loads, further optimization may be required for large-scale deployment. The accuracy of recommendations depends on the quality of data provided by users, and scam detection effectiveness can be further improved by incorporating advanced machine learning models and real-time feedback mechanisms. Nevertheless, the proposed system addresses critical gaps present in existing internship platforms.

### 7.3 Conclusion

In conclusion, SecureSmart Intern is a secure and intelligent platform that bridges the gap between students and genuine internship opportunities. By combining AI-powered internship recommendations with proactive scam detection, the system ensures a safer, more efficient, and user-centric internship search experience. The platform also assists companies in reaching suitable candidates while maintaining data security and system reliability.

The project successfully demonstrates the practical application of web technologies, database management, and artificial intelligence concepts. SecureSmart Intern can serve as a strong foundation for future enhancements and real-world deployment.

## 8 BIBLIOGRAPHIES

---

### ▪ REFERENCES

## 8.1 References

1. Internshala Official Website, <https://internshala.com>
2. LinkedIn Jobs and Internship Portal, <https://www.linkedin.com/jobs>
3. Indeed Job Search Platform, <https://www.indeed.com>
4. Tan, P. N., Steinbach, M., & Kumar, V., Introduction to Data Mining, Pearson Education
5. Aggarwal, C. C., Recommender Systems: The Textbook, Springer
6. OWASP Foundation, Web Application Security Guidelines, OWASP Documentation
7. PostgreSQL Official Documentation, <https://www.postgresql.org/docs/>
8. React.js Official Documentation, <https://react.dev>
9. Node.js Official Documentation, <https://nodejs.org>
10. Research papers on internship recommendation systems and online fraud detection from IEEE Xplore Digital Library



